

SYNOPSIS

At 1252 on 19 July 2016, two passenger carrying rigid inflatable boats, *Osprey* and *Osprey II*, collided in the Firth of Forth. A passenger who was sitting on an inflatable tube of *Osprey II* was crushed between *Osprey*'s bow and *Osprey II*'s helm console, resulting in her sustaining serious injuries.

Both RIBs had departed from Anstruther Harbour and were bound for the Isle of May. While proceeding in parallel at a speed of around 6 knots, the skipper of each RIB increased speed and commenced a power turn away from each other with the intention of passing each other in the course of completing a round turn. However, as the RIBs turned towards each other, it became apparent to both skippers that the RIBs were in danger of colliding. Although they both acted quickly to reduce the speed of their respective vessels and so lessen the impact, they were unable to prevent the collision.

The manoeuvre had previously been carried out successfully on several occasions but it had not been formally risk assessed and no thought had been given to what to do if a collision situation developed.

There are currently no regulations preventing persons on RIBs from sitting on the inflatable tubes. However, passengers not sitting on suitable inboard seating have an increased risk of falling overboard, are at significant risk of musculoskeletal injuries and, as identified in this accident, are more likely to be seriously injured in the event of a collision.

The operating company, Isle of May Boat Trips Ltd, has since banned similar manoeuvres in future and has prohibited crew and passengers from sitting on the RIB inflatable tubes.

A recommendation has been made to the Maritime and Coastguard Agency to include in its forthcoming Recreational Craft Code requirements for suitable seating in respect of commercially operated passenger carrying RIBs.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *OSPREY*, *OSPREY II* AND ACCIDENT

SHIP PARTICULARS		
Vessel's name	<i>Osprey</i>	<i>Osprey II</i>
Flag	UK	UK
Certifying Authority	Royal Yachting Association	Royal Yachting Association
Type	Humber Quinquari Rigid Inflatable Boat	Humber Quinquari Rigid Inflatable Boat
Registered owner	Isle of May Boat Trips Ltd	Isle of May Boat Trips Ltd
Manager(s)	Isle of May Boat Trips Ltd	Isle of May Boat Trips Ltd
Construction	GRP hull with hypalon inflatable tubes	GRP hull with hypalon inflatable tubes
Year of build	2004	2010
Length overall	10m	8.0m
Registered length	Not applicable	Not applicable
Gross tonnage	Not applicable	Not applicable
Coded capacity	12 passengers + 2 crew	12 passengers +2 crew
Authorised cargo	Passengers	Passengers
VOYAGE PARTICULARS		
Port of departure	Anstruther Harbour	Anstruther Harbour
Port of arrival	Isle of May (intended)	Anstruther Harbour
Type of voyage	Passenger excursion	Passenger excursion
Cargo information	Passengers	Passengers
Manning	13 (skipper +12 passengers)	12 (skipper +11 passengers)
MARINE CASUALTY INFORMATION		
Date and time	19 July 2016, 1252	
Type of marine casualty or incident	Serious Marine Casualty	
Location of incident	Firth of Forth, Scotland	
Place on board	Not applicable	
Injuries/fatalities	None	Serious injuries to one passenger

1.2 BACKGROUND

Isle of May Boat Trips Ltd owned and operated two rigid inflatable boats (RIBs), *Osprey* and *Osprey II*, from Anstruther Harbour, Firth of Forth, Scotland. The company offered passenger trips from April to September to the Isle of May (**Figure 1**), a small island in the outer Firth of Forth. The island is a haven for sea birds and has a large puffin colony. It was a popular destination for tourists and *Osprey* was permitted to land 12 passengers daily.

The company, which was owned by *Osprey*'s skipper, advertised three different RIB tours:

- *‘May Experience’ – a trip to the Isle of May with passengers disembarking onto the island for several hours before returning to Anstruther.*
- *‘Taste of May’ – a trip around the Isle of May, not landing on the Island and returning to Anstruther.*
- *‘Coastal Dash’ – a joyride along the coast to enjoy the thrill of the boat ride and experience coastline from the sea.’ [sic]*

Isle of May Boat Trips Ltd operated in partnership, and shared offices, with Anstruther Pleasure Cruises Limited, which operated the Isle of May ferry, *May Princess*. *May Princess* carried up to 100 passengers from Anstruther Harbour to the Isle of May each day from April to September.

A scheduled departure timetable was published in advance. As Anstruther Harbour was tidal, departure times varied depending on the tide. *Osprey* operated every day throughout the season, *Osprey II* did not operate on Tuesdays, Wednesdays or Fridays.

1.3 NARRATIVE

1.3.1 Events prior to departure

On Tuesday 19 July 2016, *Osprey* was fully booked with 12 passengers for its scheduled 1230 ‘May Experience’ tour departure. Fine weather was forecast and several other potential customers had expressed an interest in a trip. *Osprey II* was not scheduled to operate that day but, owing to the fine weather and high passenger demand, the decision was taken to run a ‘Taste of May’ tour.

At 0800, *Osprey*'s skipper contacted *Osprey II*'s skipper, who confirmed he was available to run the additional tour provided he could bring his 8-year old son with him. This was agreed, and throughout the morning 10 *Osprey II* spaces were sold to passengers for a 1230 departure time.

At 0930, *Osprey*'s skipper arrived at Anstruther Harbour with fuel in jerry cans for both RIBs. He then opened the booking office, and began to process bookings and paperwork for the day's trips. Soon after, *Osprey II*'s skipper arrived, accompanied by his son. Both RIBs were fuelled and then *Osprey II* was moved from its berth in the marina to the passenger boarding embarkation steps.

At 1215, the skippers distributed and fitted lifejackets to both *Osprey* and *Osprey II*'s passengers on the quayside. There were 12 passengers for *Osprey*, 11 adults and 1 child, and 11 passengers for *Osprey II*, 7 adults and 4 children, 1 being the skipper's son.

A safety briefing was given to all of the passengers by *Osprey*'s skipper. Each skipper then led his respective passengers to the boarding area and assisted with their boarding. *Osprey*'s passengers sat on bench seats configured in five rows. Eight of *Osprey II*'s passengers sat on bench seats configured in four rows, two passengers each sat in a position on an inflatable tube indicated to them by the skipper, one on each side. On *Osprey II* the skipper's son sat on the starboard inflatable tube, in a position aft of the helm console.

At 1225 *May Princess* sailed from Anstruther Harbour, bound for the Isle of May.

1.3.2 Departure from Anstruther

At 1236, *Osprey II* departed Anstruther Harbour and was joined shortly afterwards by *Osprey*. Both vessels then stopped briefly to allow photographs to be taken (**Figure 2**).

Both skippers then increased speed and proceeded towards *May Princess*, which was now 2.5 miles ahead of them. *May Princess* was travelling at around 6 knots. *Osprey II* ran parallel with and on the port side of *Osprey* at an estimated speed of 17 knots.

Osprey II then crossed behind and onto *Osprey*'s starboard side, a manoeuvre that resulted in the passenger sitting on *Osprey II*'s starboard inflatable tube being splashed.

As the RIBs approached *May Princess*'s stern, they slowed down to match its speed (**Figure 3**).

1.3.3 The collision

At 1251, both skippers glanced at each other and nodded. This was the pre-determined signal to start a turn manoeuvre. *Osprey*'s skipper increased speed and put his helm hard to port, commencing a power turn to port. *Osprey II*'s skipper increased speed and put his helm to starboard, commencing a power turn to starboard (**Figure 4**).

As the RIBs turned towards each other, it became apparent to both skippers that the RIBs were in danger of colliding. *Osprey II*'s skipper tried to avoid collision by turning to port, reducing engine speed and engaging astern gear as soon as the engine would allow. As the RIB rapidly slowed down, a child who was sitting on one of the bench seats slid off her seat and onto the floor. *Osprey*'s skipper also tried to avoid collision by turning to starboard and then to port and by engaging astern gear on both engines, which rapidly slowed the RIB (**Figure 5**).

At 1252, the passenger sitting on *Osprey II*'s starboard inflatable tube turned away from *Osprey*'s approaching bow, which then pinned her to *Osprey II*'s helm console. The starboard aft section of *Osprey II*'s inflatable tube was punctured and rapidly deflated.



Figure 2: *Osprey II* passengers shortly after departing Anstruther, showing two passengers and the skipper's son seated on the tubes

Osprey II's skipper, realising that the passenger was injured, pushed *Osprey*'s bow away, freeing her. The skipper then made an initial assessment of the injured passenger, who was in obvious pain but able to communicate. He tried to relocate her onto one of the bench seats, but she was in too much pain to move (**Figure 6**).

1.3.4 Post-collision events

At 1255, *Osprey*'s skipper reported the collision to the coastguard on very high frequency (VHF) radio Channel 16, advising that both RIBs would be returning to Anstruther Harbour with one injured passenger, and requesting the coastguard to arrange for an ambulance to meet them in Anstruther.

May Princess initially stopped following the accident, but then resumed passage towards the Isle of May once it was established that no additional assistance was required.

BS EN ISO 6185 – 4: 2011 Inflatable Boats Part 4: Boats with a hull length of between 8m and 24m with a motor power rating of 15kW and greater, is a harmonised standard referred to by the RCD and includes:

‘6.8 Seating and attachment systems (where offered as standard or optional equipment)

Seating and handholds shall provide support for spinal neutral alignment and postural stability for each person up to the crew limit and also to prevent them falling or being thrown on deck or overboard.

Buoyancy tubes shall not be used for seating areas.

7.1 Maximum permissible number of persons (crew limit)

The crew limit...shall not exceed the number of persons for which seating has been assigned...’.

Inflatable boats of over 8m in length, built to comply with the RCD after 2011 should have sufficient seating for the defined maximum permissible number of persons.

1.12 PREVIOUS/SIMILAR ACCIDENTS

Two Cardiff Bay Yacht Club RIBs (MAIB Report No 19/2011)

Two club RIBs belonging to Cardiff Bay Yacht Club collided at night while transporting a number of children across Cardiff Bay. The RIBs were proceeding at about 20 knots in the dark and carried no navigation lights or torches. One RIB had one seat and six occupants, the other, two seats and seven occupants. Those without designated seats sat on the inflatable tubes. As a result of the accident, three children, who were sitting on the inflatable tubes, were thrown into the water. One of the children sustained a traumatic brain injury, two others required subsequent extended medical treatment and several of the children from both boats suffered bruising and soreness following the collision.

Delta 8.5m RIB (MAIB Report No 1/2011)

A male passenger suffered lower back compression fractures while a RIB was transporting him and fellow workers to a jack-up barge on the River Thames. The injury occurred as the passenger landed heavily on a locker lid, where he had been sitting, after he had been momentarily lifted off the lid due to the RIB’s motion. The RIB was travelling at a speed of approximately 30 knots. There were no handholds, or foot straps in the vicinity of the locker lid. The boat had four seats and there were nine persons on board.

Celtic Pioneer (MAIB Report No 11/2009)

A female passenger on board a 9m RIB suffered a lower back wedge compression fracture. She was on a 1-hour boat trip in the Bristol Channel with 10 colleagues as part of a team building exercise. The injury occurred as the passenger was momentarily lifted into the air due to the RIB’s motion.

The MAIB report stated that the operation of boats conducting thrill rides and similar activities was not specifically included in the current MCA Code of Practice, and there was no industry approved code of practice. Consequently, the standards of safety management among UK operators of such boats varied considerably.

Milly (MAIB Report No 5/2014)

Six people were ejected from a RIB in the Camel Estuary. The ejection occurred as the vessel hooked during a high speed turn. The RIB continued to circle, striking several persons in the water resulting in two fatalities and seriously injuring two others. The driver was not wearing an engine kill cord, and not all of the occupants were sitting on designated seats.

E.R.Athina (MAIB Report No 3/2013)

The bosun on board the Liberian registered platform supply ship *E.R.Athina* was crushed between the ship's hull and the lifting frame of the ship's fast rescue craft (FRC). The bosun was standing at the stern of the FRC, and was trying to push the boat away from the ship's hull as a repair was being made to a small area of damaged paintwork on the hull.

Following the accident the bosun was examined on board the ship. However, because his injuries were internal and he was conscious and responsive, the potential severity of his condition was not immediately apparent. The bosun suffered severe internal chest injuries and was evacuated ashore for medical treatment. He died soon after arriving at hospital.

Kinghorn Inshore Lifeboat (MAIB Preliminary Examination July 2007)

Kinghorn ILB, a 7.3m RIB, was conducting a routine exercise in fine, calm weather, and at the time of the accident it was carrying out a series of turns at speeds of between 20 and 25 knots.

A crew member, seated on the port side inflatable tube, fell overboard during a turn to starboard. He was struck on the head by the RIB's engine propeller. Despite wearing a helmet, the crewmember suffered severe head injuries.

The RNLI's own internal investigation concluded that all crew must be properly seated and secure when practising high speed turns.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 ACCIDENT OVERVIEW

This accident occurred when two passenger tour RIBs collided during a manoeuvre designed to make the trip more exciting. The passengers on board each RIB were not expecting such a manoeuvre as neither trip was advertised or sold as a thrill ride. The analysis focuses on the power turn manoeuvre and the vulnerability of passengers to injury.

2.3 THE POWER TURN MANOEUVRE

Osprey and *Osprey II* collided as a result of turning towards each other at close range (**Figure 11**). The power turn manoeuvre had previously been carried out successfully on several occasions but it had not been formally risk assessed and no thought had been given to what to do if a potential collision situation developed.

This was the first time that the power turn manoeuvre had been carried out with *Osprey II* initially positioned on the starboard side of *Osprey*. On all previous occasions, the turn had been started with *Osprey II* on the port side of *Osprey*, the RIBs having crossed each other's path twice prior to the manoeuvre. On this occasion, the RIBs had crossed only once on account of the passenger sitting on *Osprey II*'s starboard inflatable tube getting splashed during the first crossing. The power turn therefore commenced with the RIBs in a different orientation and being turned in the opposite direction to normal.

Both RIBs were travelling at approximately 6 knots when they increased speed and executed the power turn. When it became apparent to both skippers that the power turn had resulted in the potential for a collision, both acted quickly to reduce the speed of their respective vessels and so lessen the impact. However, due to the closing speed and the lack of sea room, the skipper had insufficient time in which to prevent a collision.

The fact that the power turn had previously resulted in the RIBs passing each other starboard side to starboard side may explain why *Osprey II*'s skipper instinctively turned to port. Had more consideration been given to the manoeuvre when it was first conceived, it would have been possible to agree how the RIBs were to pass each other in any of the possible starting configurations.

2.4 SEATING ARRANGEMENTS

Driving RIBs at speed through waves and across the wake of other craft, commonly referred to as 'thrill rides', can be carried out safely providing all of the risks are properly identified and mitigated. Interacting with other vessels brings additional risks.

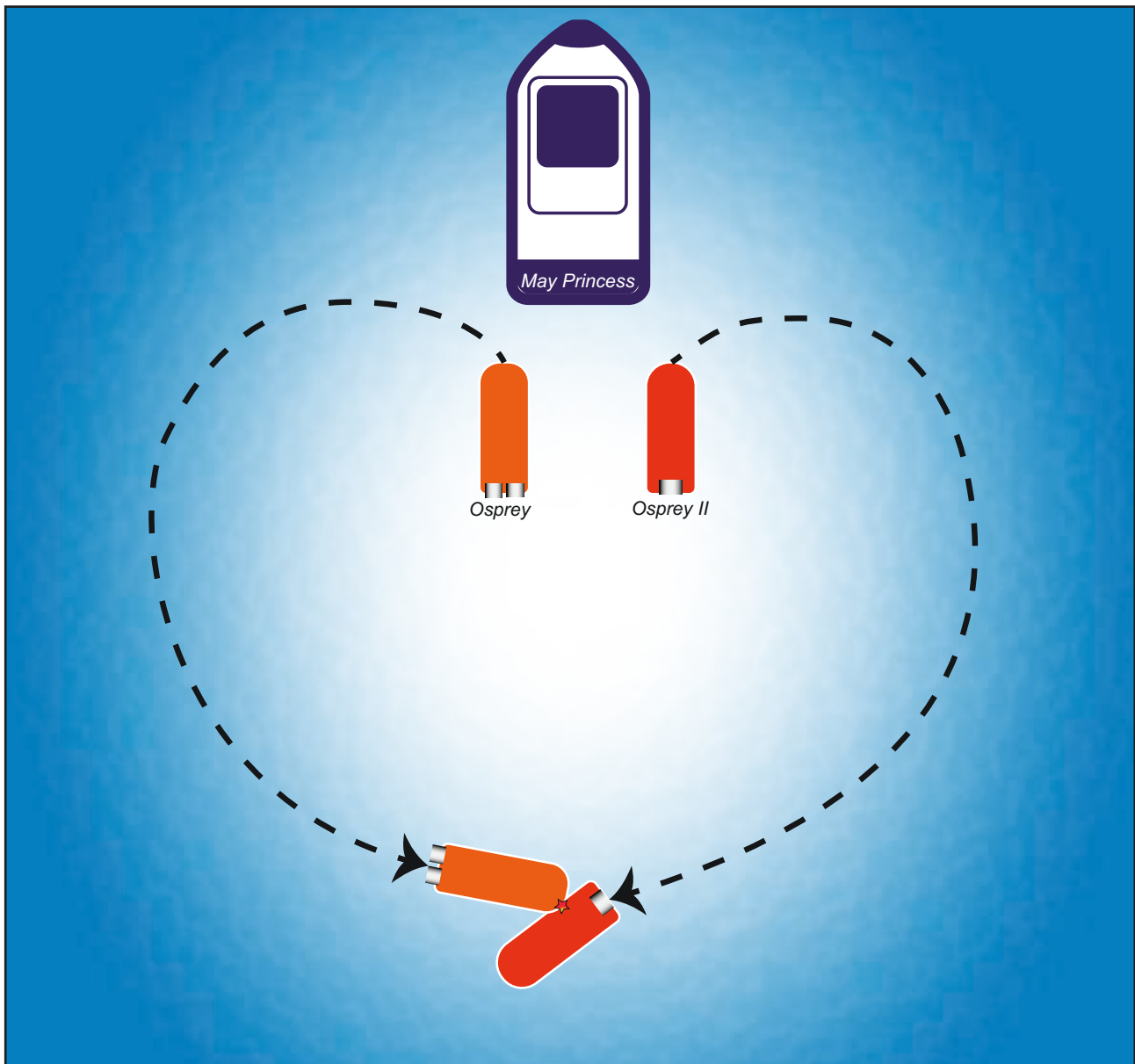


Figure 11: Power turn manoeuvre

Isle of May Boat Trips Ltd had not advertised the 'May Experience' and Taste of May' tours as 'thrill rides'. However, the skippers had decided to incorporate a level of excitement into the trips when *Osprey* and *Osprey II* were operating together. The additional risks associated with these manoeuvres had not been recognised and so the guidance contained in MGN 436 (M+F) with respect to passenger posture and speed perception was not implemented.

This investigation and many of the MAIB reports listed in section 1.12 concern accidents where passengers were not provided with an appropriate seat. The safety issues arising from these investigations highlight the need for effective risk assessment to determine suitable passenger seating for the intended activity.

Although *Osprey II* was certified to carry 12 passengers, including 4 passengers in designated positions on the inflatable tubes, Isle of May Boat Trips Ltd had limited passenger numbers in other than good weather to 8, the maximum number that could be accommodated on the RIB's bench seating. In good weather, as on the day of the accident, it allowed only 2 additional passengers to be carried on the inflatable

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The power turn manoeuvre had previously been carried out successfully on several occasions but it had not been formally risk assessed and no thought had been given to what to do if a potential collision situation developed. [2.3]
2. The power turn commenced with the RIBs in a different orientation and being turned in the opposite direction to normal. Had more consideration been given to the manoeuvre when it was first conceived, it would have been possible to agree how the RIBs were to pass each other in any of the possible starting configurations. [2.3]
3. Due to the closing speed and the lack of sea room, the skippers had insufficient time in which to react and prevent a collision. [2.3]
4. There are currently no regulations preventing persons on RIBs from sitting on the inflatable tubes. [2.4]
5. Passengers not sitting on suitable inboard seating have an increased risk of falling overboard, are at significant risk of musculoskeletal injuries and, as demonstrated by this accident, are exposed to serious injury in the event of a collision. [2.4]
6. Current regulation and guidance provide no metrics in terms of vessel speed or weather conditions in relation to what constitutes suitable seating, and make no specific reference to other factors that may need to be taken into account to reduce the risk of serious injury in the event of a collision. [2.5]
7. Of particular concern is the absence of a common understanding between operators, the MCA, CAs, passengers and port authorities as to when seating on a RIB's inflatable tubes is considered to be unsuitable. [2.5]

3.2 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The decision to additionally carry *Osprey II*'s skipper's son resulted in the company's self-imposed passenger number limit being exceeded and in the skipper's son sitting in an undesignated seat. [2.4]

3.3 OTHER SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT²

1. The additional risk associated with the thrill seeking activity had not been recognised and so current published guidance with respect to passenger posture and speed perception was not implemented. [2.4]

² These safety issues identify lessons to be learned. They do not merit a safety recommendation based on this investigation alone. However, they may be used for analysing trends in marine accidents or in support of a future safety recommendation.

2. *Osprey II*'s skipper did not have his engine cut-out kill cord attached when the RIB departed Anstruther Harbour on the day of the accident. [2.6]
3. Internal injuries are hard to diagnose without specialist medical expertise. Unless the extent of an injury is clear and can be competently addressed by those present, appropriate medical assistance should be sought immediately. [2.7]